MIL-P-52539B(ME)
11 July 1984
SUPERSEDING
MIL-P-52539A(ME)
10 March 1975

MILITARY SPECIFICATION

PIPELINE WELD SET, TESTING

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pipeline weld testing set consisting of a coupon cutting machine and a coupon testing machine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6635

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-8788	- Hose, Hydraulic, High Pressure.
MIL-F-8789	- Fitting End, Attachable Hydraulic, High Pressure Hose.
MIL-T-13880	- Torches, Oxyacetylene Welding and Cutting.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN PETROLEUM INSTITUTE (API)

API 5L - Line Pipe.
API 1104 - Welding Pipe Lines and Related Facilities.

(Application for copies should be addressed to the American Petroleum Institute Publications and Distribution Section, 2101 L Street Northwest, Washington, D.C. 20037).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pipeline weld testing set (hereinafter referred to as the weld testing set) shall consist of a coupon (test specimen) cutting machine and a coupon tensile and guided-bend testing machine.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subjected to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The weld testing set shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Coupon cutting machine. The coupon cutting machine shall consist of a frame, a holder for a straight type oxgen-acetylene torch conforming to MIL-T-13880, type II with optional cutting attachment, and feed screws. The frame shall be high-strength aluminum or steel tube to hold the torch in the correct position and alinement while cutting; the holder shall be provided for positioning of the torch. The coupon cutting machine shall cut coupons as specified in API 1104.

3.4.1 Feed screws. Movement of the torch holder and torch shall be controlled by feed screws operated by handles designed to allow continuous motion of the torch. The feed screws shall have acme or square threads as specified in FED-STD-H28.

3.5 Coupon testing machine. The coupon testing machine shall consist of a hand-operated hydraulic pump and indicating gage, bending jigs, tension specimen holder, hydraulic ram, hose, and supporting frame and structure. The machine shall be capable of conducting the tensile test and guided-bend test specified in API 1104 on the following sizes and wall thicknesses of pipe conforming to API 5L without permanent distortion:

Nominal pipe size	Wall thickness
8-inch	0.375-inch
12-inch	0.250-inch
16-inch	0.312-inch
20-inch	0.500-inch

3.5.1 Hydraulic system. A reservoir with a capacity for extreme movement of the hydraulic ram shall be integral with the hydraulic pump. The hydraulic fluid supply system shall be filled with the fluid recommended by the contractor. Provisions shall be made for draining the hydraulic fluid. The hydraulic system shall not leak.

3.5.2 Indicating gage. The gage shall be calibrated to indicate the load, in thousands of pounds applied to the coupon, and shall be capable of measuring the loads required to conduct the tests specified in 3.5. The gage dial shall be not less than 3-1/2 inches or more than 6 inches in diameter and shall be mounted in a position to be read easily by the operator of the machine. The gage shall have an indicating hand to indicate the exact load being applied, and an indicating hand that remains at the point of maximum load when the pressure is released. The gage shall be marked in red on the high end of the dial to indicate ram overload pressure. The gage shall be equipped with a manual device to return the indicating hand to zero at the end of each test.

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3.5.3 Bending jigs. The bending jigs shall be in accordance with API 1104 for the size of pipe specified in 3.5.

3.5.4 Coupon holder. The coupon holder shall be a wedge jaw design that increases its grip on the coupon as the load on the coupon is increased. The jaws and jaw holders shall be designed to permit easy removal of the jaws and coupons on completion of a test.

3.5.5 Hydraulic ram. The hydraulic ram shall be large enough to develop the loads and amount of movement required. The ram head shall permit easy rapid changing of the coupon holder and the bending jigs.

3.5.6 Hose. The hose shall be a high-pressure hydraulic oil hose not less than 4 feet long conforming to MIL-H-8788. Hose connections shall be leakproof. Hose fittings shall conform to MIL-F-8789.

3.5.7 Supporting frame and structure. The supporting frame and structure shall consist of the hand-operated hydraulic pump mounting and the hydraulic ram mounting.

3.5.7.1 Hand-operated hydraulic pump mounting. The pump mounting shall provide a solid foundation for hand operation of the hydraulic pump when set on solid, level ground. The hydraulic pump shall be mounted so that the operating handle is pushed downward against the mounting to increase hydraulic pressure.

3.5.7.2 Hydraulic ram mounting. The hydraulic ram mounting shall include a mounting for the pulling jaws for the tensile tests, and jigs for the guided-bend tests. The mounting shall provide for easy removal and replacement of the pulling jaws and the bending jigs.

3.6 Safety. The weld test set shall not present hazards to the operator or maintenance and repair personnel.

3.7 Human factors engineering. The weld test set shall conform with accepted criteria for design for human factors as specified in MIL-STD-1472.

3.8 Storage boxes. Separate steel boxes shall provide storage and protection for the coupon cutting machine and its components, and for the coupon testing machine and its components.

3.9 Identification marking. The weld testing set shall be identified in accordance with MIL-STD-130. The marking shall be applied to the weld testing set on metal plates attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

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3.10 Treatment and painting. The portions of the weld testing set normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.11 Workmanship.

3.11.1 Castings and forgings. All parts, components, and assemblies of the testing set which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.11.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flamecutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The first article weld testing set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article weld testing set shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 4.0 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The weld testing sets shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Identification marking missing or illegible.
106. Treatment and painting not as specified.
107. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Coupon cutting machine. Operate the coupon cutting machine to cut 4 coupons from each pipe diameter and wall thickness specified in 3.5. Nonconformance to 3.4 shall constitute failure of this test.

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4.5.2.2 Coupon testing machine. Using the coupons from 4.5.2.1, perform the tensile and guided-bend tests as specified in API 1104. The welding test set shall be examined for the following four conditions and the failure of any condition(s) shall constitute failure of this test.

a. Permanent distortion of stressed parts.

b. Leaking hydraulic fluid.

c. Inability of the machine to develop the tensile or bending stresses required.

d. Coupon gripping jaws flying out of the holder.

4.5.2.3 Safety. The weld test set shall be evaluated for safety requirements throughout testing as specified in 4.5.2. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.4 Human factors. The human factors requirements shall be evaluated during testing. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials and containers not as specified. Each incorrect material or container shall be considered one defect.
109.	Disassembly not as specified for level A.
110.	Surfaces of components requiring a contact preservative in accordance with the referenced document not coated as specified for level A.
111.	Feed screws not preserved as specified for level A.
112.	Indicating gage not protected as specified for level A.
113.	Components not placed in their storage boxes as specified for level A and commercial.
114.	Marking illegible, incorrect, incomplete, or missing for level A, B, or commercial.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Disassembly. Disassembly of the weld testing set shall be the minimum necessary to safeguard parts known to be subject to damage or loss in accordance with the manufacturer's recommendations, or to effect reduction of cubage. Bolts, nuts, washers, and pins shall be placed in one of the mating parts and secured to prevent their loss. Torch holder, feed screws, gage, bending jigs, tension holder, or any other component removed to prevent damage, deterioration, or excessive cubage, shall be secured to that unit from which removed.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated as follows:

5.1.1.3.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.3.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable with barrier material conforming to MIL-B-121, type I or II, grade A, class 2. The material shall be secured in place with tape conforming to MIL-T-22085, type II.

5.1.1.4 Coupon cutting machine. The feed screws shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I or II, grade A, class 2, extending the full length of the screws and secured with tape conforming to MIL-T-22085, type II. The coupon cutting machine and components shall be placed in the storage box provided. The contents shall be blocked and braced to prevent movement or damage.

5.1.1.5 Coupon testing machine. The hydraulic fluid supply system shall be filled as specified in 3.5.1. Hoses shall not be removed or disconnected. The indicating gage face shall be covered with fiberboard and the fiberboard secured with tape conforming to MIL-T-22085, type II. The coupon testing machine and components shall be placed in the storage box provided. The contents shall be blocked and braced to prevent movement or damage.

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5.1.2 Commercial. The coupon cutting machine and components, and the coupon testing machine and components shall be preserved in accordance with ASTM D 3951 and placed in their storage boxes.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The components comprising a complete weld testing set shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. Blocking and bracing to prevent movement or damage shall be provided. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. The weld testing set shall be packed as specified in 5.2.1 for level A, except the box shall be domestic type or class as applicable.

5.2.3 Commercial. The weld testing set, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951; in addition, weight and cube of the pack shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The weld testing set is intended for use in the field for cutting and testing pipe weld coupons and for conducting welding procedure evaluation tests and qualification tests of welders.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Color required (see 3.10).

d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item shall be examined and should be a first produced weld testing set. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each weld testing set.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of weld testing sets.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52539B(ME)	Pipeline Weld Set, Testing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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